

11.17 Classwork: Cumulative Frequency and Normal Models

Answer on lined paper. Problem 1 is no calculator. Problem 2 is calculator allowed.

1. [Maximum mark: 17]

AA SL 2023 Nov P1 TZ2 Q7

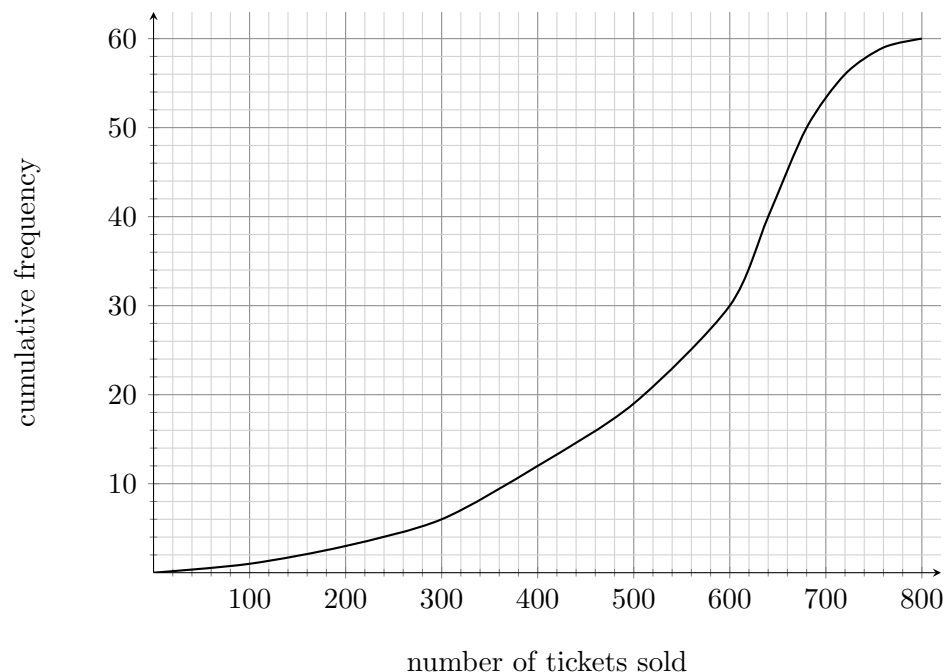
A ballet company performs *The Nutcracker* every year. Last year they gave a total of 60 performances at their theatre which has a maximum capacity of 800. The number of tickets sold, n , at each performance is shown in the following frequency table.

Number of tickets sold, n	Number of performances
$0 < n \leq 200$	3
$200 < n \leq 400$	p
$400 < n \leq 600$	18
$600 < n \leq 800$	30

(a) (i) Find the value of p . [1]

(ii) Write down the modal class. [1]

The following cumulative frequency diagram also displays these data.



Use the cumulative frequency curve to estimate

(b) (i) the median number of tickets sold; [1]

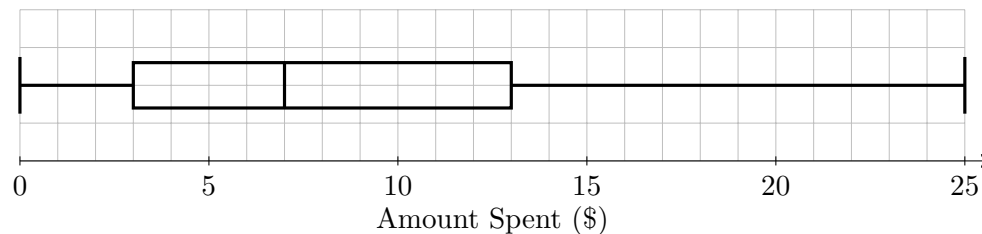
(ii) the number of performances where at least 80% of the tickets were sold. [3]

After a performance, the company decides to conduct a survey to obtain feedback from the audience.

- (c) (i) State one disadvantage of the company surveying only the first 5% of the audience as they leave the theatre. [1]
- (ii) Describe briefly how the company could collect feedback from 5% of the audience using the systematic sampling method. [2]
- (iii) State the sampling method which should be used if the survey is to be representative of the number of children and the number of adults in the audience. [1]

Last year 36 000 tickets were sold to *The Nutcracker*.

The following box and whisker diagram displays the amount spent by the audience at the souvenir shop when they attended the performance.



- (d) (i) Estimate the number of people who spent between \$3 and \$25. [2]
- (ii) Half the audience spent less than \$ a . Estimate the value of a . [1]

This year the company will again give 60 performances and expects to sell 17 additional tickets for each performance.

- (e) (i) Calculate the mean number of tickets the company expects to sell this year for each performance. [3]
- (ii) State what effect, if any, this increase in ticket sales would have on the variance of the number of tickets sold for each performance. [1]

2. [Maximum mark: 16]

AA SL 2023 Nov P2 TZ1 Q9

A farmer is growing a field of rice plants. The height, H cm, of each plant can be modelled by a normal distribution with mean μ and standard deviation σ .

It is known that

$$P(H < 82.4) = 0.213 \quad \text{and} \quad P(H > 87.3) = 0.409.$$

(a) Find the probability that the height of a randomly selected plant is between 82.4 cm and 87.3 cm. [2]

(b) Find the value of μ and the value of σ . [5]

The farmer measures 100 randomly selected plants. Any plant with a height greater than 87.3 cm is considered ready to harvest. Heights of plants are independent of each other.

(c) (i) Find the probability that exactly 32 plants are ready to harvest. [2]

(ii) Given that fewer than 44 plants are ready to harvest, find the probability that exactly 32 plants are ready to harvest. [4]

In another field, the farmer is growing the same variety of rice, but is using a different fertilizer. The heights of these plants, F cm, are normally distributed with mean 92.8 and standard deviation d . The farmer finds the interquartile range to be 4.52 cm.

(d) Find the value of d . [3]